

Core Compulsory Readings

1. Free online introduction to LaTeX Available at [https://www.overleaf.com/learn/latex/Free_online_introduction_to_LaTeX_\(part_1\)](https://www.overleaf.com/learn/latex/Free_online_introduction_to_LaTeX_(part_1))
2. Learning Latex, D.F.Griffits, D.J.Higham, Siam, Philadelphia, 1997
3. Latex, <https://en.wikibooks.org/wiki/LaTeX>

Core Suggested Readings

1. Online materials related to document preparation using Latex and slide preparation using Beamer

TEACHING LEARNING STRATEGIES

- Lecturing, Demonstration.

MODE OF TRANSACTION

- Lecture, Seminar, Discussion, Notes

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 3 credit courses with 2 Credit Theory + 1 Credit Practical.

KU5SECCSE204 Data Processing with Python

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	SEC	200	KU5SECCSE204	3	60

Learning Approach (Hours/ Week)	Marks Distribution	
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Lecture	Practical/ Internship	Tutorial	CE	ESE	Total	Duration of ESE (Hours)
2	2	0	50	50	100	2(T)+3(P)*

* ESE Duration: 2 hours for theory and 3 hours for Lab

Course Description: This course includes an overview of the various tools available for writing codes and running the same in Python, and gets students coding quickly. It also provides hands-on coding exercises using commonly used data structures and writing custom functions after preprocessing the dataset.

Course Objectives:

- Understand how to perform some basic tasks to start exploring and analyzing the imported data set
- Learn how to perform computations on the data to calculate basic descriptive statistical information
- Learn how to manipulate data efficiently.
- Understand the visualization and analytics of data.

Course Outcome

At the end of the Course, the Student will be able to:

SL #	Course Outcomes
CO1	To understand python fundamentals and data types used in python
CO2	To learn basic statistical measurements required for machine learning
CO3	To familiarize preprocessing of data before processing.
CO4	To experience visualization of 1D and 2D data

Mapping of COs to PSOs

CO - PSO Mapping					
	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	✓	✓	✓	✓	✓
CO2	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓

COURSE CONTENTS

Module 1: Python Fundamentals: Basic Concepts, Naming Variables, Operators and operands, Expressions, importing libraries, Redirecting the output, Data types, Lists, Working with files, Working with Directories; Big Data vs Data Science **(15 Hours)**

Module 2: Introduction to probability: discrete and continuous variables, probability distribution, Bayes Rule, Variance, Standard Deviation, Types of probability distributions, Algorithms designed using probability, **(15 Hours)**

Module 3: Finding datasets, Jupiter notebooks and loading data, pandas vs numpy, Saving, Dropping Null Values, User Defined Functions, Cleaning Dataset, Graphs and Statistics, Histogram, Working with rows and columns. **(15 Hours)**

Module 4: Introduction to basic plots, pandas vs matplotlib, visualizing 1D distributions, visualizing 2D distributions, Higher dimension visualizations. **(15 Hours)**

Core Compulsory Readings

1. Wes McKinney, "Python for Data Analysis", 2nd Edition, O'Reilly
2. Joel Grus," Data Science from Scratch" First Edition, April 2015

3. McKinney, W. (2012). Python for data analysis: Data wrangling with Pandas, NumPy, and IPython. " O'Reilly Media, Inc."

TEACHING LEARNING STRATEGIES

Lecturing, case study/mini projects, Team Learning, presenting seminars on selected topics, Digital Learning

MODE OF TRANSACTION

Lecture, Seminar, Discussion, Demonstration, Questioning and Answering, Audio, Video, Print.

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 3 credit courses with 2 Credit Theory + 1 Credit Practical.

Sample Questions to test Outcomes

1. What skills are necessary for a Data Scientist?
2. Statistics
3. Deep Learning
4. Linear Algebra
5. All of the above
6. CLI stands for
7. Command Line Interface
8. Command Language Interface
9. Command Line intercom
10. None of the Above
11. Give one example where both false positives and false negatives are important equally?
12. Write the difference between data analytics and data science.
13. What are data science tools?
14. List different data types available in python with examples for each.